

Step 1 of the scope-completion roadmap (v1.1): certification of the

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Step 1 certifies the natural boundaries of the Reading-H operating window, replacing the single operating point $(I, \mu^2) = (2 \times 10^{-3}, 0.005)$ by an explicit certified region. This widens the *parameter* scope; it does NOT widen the competitor class (Step 2, EXT). No tier change. Operator verdict 2026-06-10: ACCEPTED AS STEP 1 CLOSURE.

2. Two boundaries; the selection boundary binds

(A) off-diagonal: $R_{\text{lead}} = 23.2(1 + K_{\text{floor}})I < 1 \Rightarrow I < I_c^{\text{off}} = \frac{1}{23.2 \cdot 14} = 3.08 \times 10^{-3} \text{ (} K_{\text{floor}} \leq 13 \text{)},$

(S) selection: $\text{joint}(\mu^2, I) > 1, \quad \text{joint} = \frac{\text{MARGIN}}{D}, \quad D = (1 + R_{\text{max}})C_2 + \frac{\text{MARGIN} - C_2}{S_{\text{un}}}.$

Numerically $I_c^{\text{sel}} = 2.41 \times 10^{-3} < I_c^{\text{off}} = 3.08 \times 10^{-3}$ (both $\times 10^{-3}$), so the **selection boundary binds**, and

$$I_c = \min_{\mu^2 \in [\times 0.5, \times 2]} I_c^{\text{sel}}(\mu^2) = 2.41 \times 10^{-3}$$

(at the low- μ^2 edge, I_c^{sel} being increasing in μ^2). The operating endpoint $I = 2 \times 10^{-3}$ is interior with headroom $(I_c - I)/I = 20\%$. I_c is a PHYSICAL selection phase boundary (joint $\rightarrow 1$ = third-cumulant degeneracy), not a proof gap.

3. The widened μ^2 window: to the physical branch boundary

$\text{MARGIN}(\mu^2) = \text{PB}(M_+) - \text{DIP_BAND}$ requires the upper PD branch $M_+ = (-3U + \sqrt{9U^2 - 60V\mu^2})/(30V)$, i.e. $\text{disc} = 9U^2 - 60V\mu^2 > 0$:

$$\mu^2 < \mu_{\text{max}}^2 = \frac{3U^2}{20V} = 0.0342.$$

Across $(0, \mu_{\text{max}}^2)$: $\text{MARGIN} > 0$, $\text{joint}(\mu^2, 2 \times 10^{-3}) > 1$, branch correct; the band is widened from $[\times 0.5, \times 2]$ to $(\times 0, \times 6.85)$, its ceiling a physical branch boundary ($\text{disc} = 0$, M_+ ceases to exist), not a proof gap.

4. Monotonicity: analytic + interval-certified (operator review 2026-06-10)

The window claims rest on three monotonicity facts, here promoted from a 3-point grid to analytic + 400-point interval certificates (res5_032 v1.1).

(M1) $\text{MARGIN}(\mu^2)$ **strictly increasing**. $\text{DIP_BAND} = |\hat{r}_0|^{3/2}/(3\sqrt{V})$ with $\hat{r}_0 = \mu^2 - 3U^2/(20V) < 0$ on the range, so $|\hat{r}_0| = 3U^2/(20V) - \mu^2$ is *analytically* strictly decreasing $\Rightarrow$ DIP_BAND strictly decreasing; $\text{PB}(M_+)$ is interval-certified strictly increasing (400-pt grid, min slope $d\text{PB}/d\mu^2$ giving $d\text{MARGIN}/d\mu^2 \geq 0.095 > 0$). Hence $\text{MARGIN} = \text{PB} - \text{DIP_BAND}$ strictly increases (both terms favourable).

(M2) joint **strictly decreasing in I (analytic)**. With $\rho = \text{MARGIN}/C_2$ and $\text{composed} = \text{MARGIN}(1 - 1/\rho) = \text{MARGIN} - C_2$,

$$D(C_2) = (1 + R_{\max})C_2 + \frac{\text{MARGIN} - C_2}{S_{\text{un}}} = C_2 \left(1 + R_{\max} - \frac{1}{S_{\text{un}}}\right) + \frac{\text{MARGIN}}{S_{\text{un}}}, \quad 1 + R_{\max} - \frac{1}{S_{\text{un}}} = 0.500 > 0,$$

so D is strictly increasing in C_2 ; and $C_2 = (\lambda I)^2 J_{\text{eff}} n_{\text{pack}}/(4(1 - a_0))$ is strictly increasing in I (the I^2 factor; interval-certified). Therefore $\text{joint} = \text{MARGIN}/D$ is strictly decreasing in I , and I_c^{sel} is the unique crossing joint = 1.

(M3) $I_c^{\text{sel}}(\mu^2)$ **strictly increasing**. joint is interval-certified strictly increasing in μ^2 ; with (M2) the implicit function theorem gives $dI_c/d\mu^2 = -(\partial_{\mu^2}\text{joint})/(\partial_I\text{joint}) > 0$ (numerator > 0 , denominator < 0); verified on a dense μ^2 grid. Hence the binding I_c sits at the low- μ^2 edge, as used in Sec. 2.

5. The certified operating region

$$\mathcal{R} = \{(I, \mu^2) : \mu^2 \in (0, \mu_{\max}^2), I \in (0, I_c^{\text{sel}}(\mu^2))\}, \quad I_c^{\text{sel}} \text{ strictly increasing (M3),}$$

operating point $(2 \times 10^{-3}, 0.005)$ interior. The enacted selection is stable throughout $\mathcal{R}$; its boundaries I_c^{sel} and $\mu_{\max}^2$ are physical.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) “*Is monotonicity a mere grid scan?*” ADDRESSED in v1.1: (M1) DIP_BAND decreasing is analytic, PB increasing is interval-certified (400 pts, min slope > 0.05); (M2) joint decreasing in I is analytic (denominator coefficient $0.500 > 0$, $C_2 \sim I^2$); (M3) I_c^{sel} increasing is the implicit function theorem from (M1)/(M2). Grade CERTIFIED WINDOW BOUNDARY, not merely EXECUTED.

(β) “*Does this widen the T7?*” No — it certifies the parameter-window boundaries; the competitor class is unchanged (Step 2, EXT). No tier change.

(γ) “*Could $I_c^{\text{off}} < I_c^{\text{sel}}$ under a tighter K_{floor} ?*” For the lattice worst $K_{\text{floor}} = 13$, $I_c^{\text{off}} = 3.08 \times 10^{-3} > I_c^{\text{sel}}$; a smaller K_{floor} only raises I_c^{off} , so selection remains binding.

Quantitative sanity check (mandatory): *intensity* $I_c = 2.41 \times 10^{-3} < I_c^{\text{off}} = 3.08 \times 10^{-3}$ (both $\times 10^{-3}$); endpoint 2×10^{-3} interior (headroom 20%); *mass* $\mu_{\max}^2 = 3U^2/(20V) = 0.0342$, disc = 0 there; *monotonicity* $d\text{MARGIN}/d\mu^2 \geq 0.095 > 0$, denom coefficient $0.500 > 0$. All from res5_032 v1.1 (6/6).

7. Result footer

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Result ID:          B1/B2 operating-window certification (roadmap
Step 1, v1.1)
Precise statement:  Binding I_c = 2.41e-3 (SC-SCOPE selection
                    joint=1, low-mu^2 edge;
                    off-diagonal R_lead=1 at 3.08e-3 looser --
                    2.41e-3 < 3.08e-3 -- so
                    selection binds); endpoint I=2e-3 interior, 20%
                    headroom. mu^2 window
                    (0, mu^2_max=3U^2/(20V)=0.0342), ceiling =
```

physical disc=0 PD branch
boundary. Monotonicity ANALYTIC+INTERVAL:

MARGIN
inc
(DIP_BAND
analytic

dec + PB 400-pt interval inc, slope>=0.095);
joint dec in I (analytic,
denom coeff 1+RMAX-1/SUNSET=0.500>0, C2-I^2);
I_c^sel inc in mu^2 (IFT).
Certified region R = {mu^2 in (0,mu^2_max), I
in (0,I_c^sel(mu^2))}.

Scope: Parameter-window boundaries of the enacted
scope. Both boundaries
physical. No competitor-class widening (Step 2
EXT). No tier change.

Dependencies: scscope_endpoint_sweep.joint;
sectorb_common.margin_of / J_of_t;
R-029 (SC-SCOPE critical I); T-018 (R_lead/K*
identity).

Evidence grade: CERTIFIED WINDOW BOUNDARY (analytic +
interval) + EXECUTED
(res5_032_window_certification.py v1.1, 6/6).

Reproduction command: python
codes/vacuum/res5_032_window_certification.py

Expected output: I_c=2.41e-3 (selection binds < 3.08e-3);
MARGIN min slope 0.0946;
denom coeff 0.500; mu^2_max=0.0342; 6/6 PASS.

Falsifier: joint>1 failing inside R; margin<=0 or
branch_ok=False for some mu^2
in (0,mu^2_max); MARGIN or joint non-monotone
on a finer grid; or
I_c^off < I_c^sel under a tighter K_floor.

Tier before / after: No change. B1/B2 enacted scope-qualified T7.
Window boundaries
CERTIFIED (analytic+interval), operator
ACCEPTED AS STEP 1 CLOSURE.

No-overclaim statement: Certifies the parameter-window of the enacted
scope; does NOT widen
the competitor class and does NOT change tier.
Global/unconditional
Reading-H T7 awaits Step 2 (EXT,
competitor-class widening to C_full).

Next required action: Step 2 -- EXT additive-energy extremal theorem
(lattice -> C_full).